

Beyond antibodies in GAD65 autoimmunity: Reframing pathogenesis through intrathecal B–T cell cross-talk

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One of the most important developments in neuroimmunology over the last decades has been the identification and characterization of a growing spectrum of autoantibody-associated neurological diseases. This progress has fundamentally reshaped the field by identifying autoimmune encephalitides and related syndromes as distinct, mechanistically diverse disease entities rather than poorly defined inflammatory disorders. This has not only improved diagnostic precision but has also provided major insights into pathogenesis, clinical phenotyping, and treatment stratification. At the same time, these advances have highlighted that antibody positivity alone does not necessarily equate to a purely antibody-mediated pathophysiology, underscoring the need to define the broader cellular and immunological context in which these syndromes arise.

One of those diseases are GAD65-related inflammatory disorders (1, 2). GAD65 syndromes are often chronic, partially treatment-responsive, and phenotypically diverse, especially across the phenotypes of limbic encephalitis/temporal lobe epilepsy, cerebellar ataxia, and stiff-person spectrum disorders (1). Clinical neuroimmunology is challenged with a central, unsolved question in anti-GAD65 neurological autoimmunity: What is the relevant pathophysiological mechanism driving

the disease and is the release of autoantibodies, regarded as prerequisite to confirm a pathogenic link with GAD autoimmunity (2), a marker of a broader, antigen-driven intrathecal immune response? And to what extent are T cells involved in the process?

By combining single-cell transcriptomics with paired immune-repertoire analysis in CSF and blood, Barman et al. argue for a compartmentalized, chronic CNS immune process centered on clonally expanded CD4⁺ stem cell-like memory T cells (TSCM), alongside somatically matured GAD65-reactive

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Competing interest statement: The authors cooperated with Kyverna in individual healing attempts in anti-GAD65 SPS. R.G. serves on scientific advisory boards with Kyverna. The authors declare no further conflict of interest in relation to the content of this manuscript.

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See companion article, "Immune cell profiling reveals expanded stem cell-like memory T cells in anti-GAD65-associated neurological syndromes," [10.1073/pnas.2514753123](https://doi.org/10.1073/pnas.2514753123).

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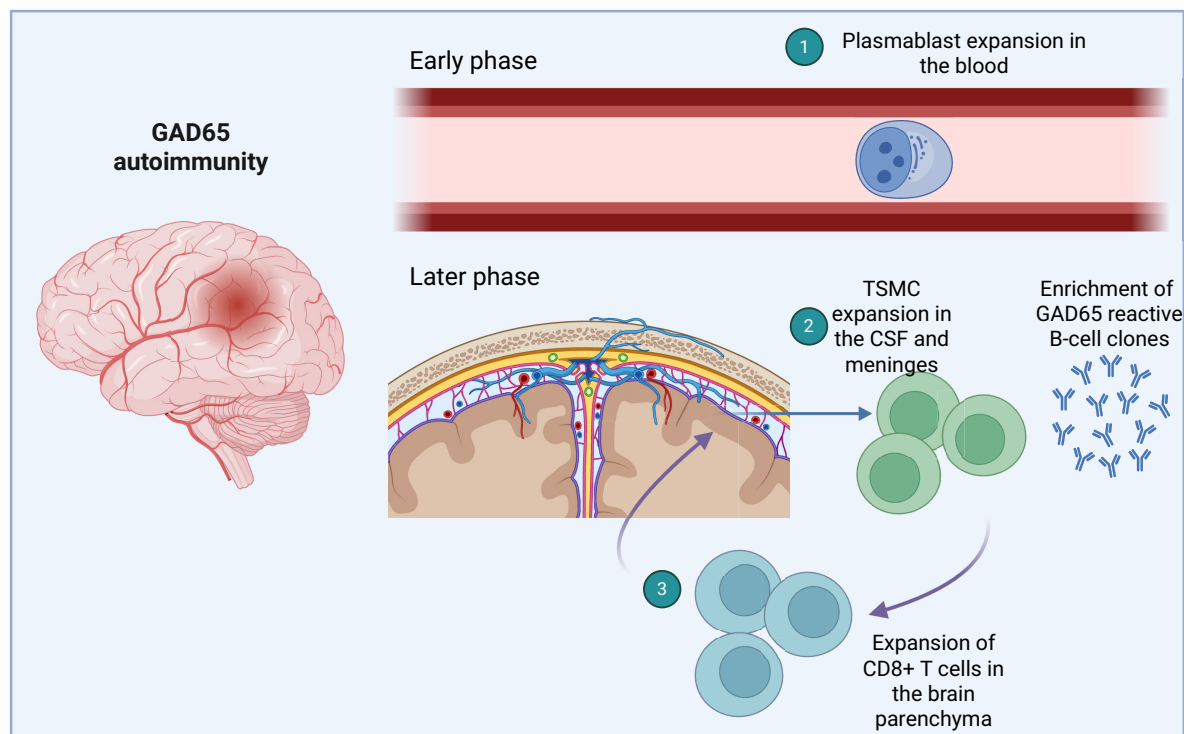


Fig. 1. Pathophysiological concept of GAD65 autoimmunity as evidenced by the findings from Barman et al. Created in BioRender. Faissner, S. (2026) <https://biorender.com/i2ie2b1>.

B cells in the CSF. The authors profiled CSF and peripheral blood from largely treatment-naïve anti-GAD65 autoimmune neurological syndrome (AINS) patients (8 for CSF, 19 for blood) and different noninflammatory or healthy controls (8 intracranial idiopathic hypertension for CSF, 20 healthy controls for blood), generating more than 118,000 single-cell transcriptomes complemented with recombinant monoclonal antibody production from CSF BCR sequences. The most striking observation is the enrichment of an immune cell cluster of TSCM in CSF, particularly in patients with longer disease duration, while plasmablast expansion is more prominent in blood early in the course of disease. Those CSF TSCM were clonally expanded and showed a proinflammatory gene signature. The temporal separation is highly interesting and biologically plausible, suggesting a model in which an early humoral burst may seed disease, while chronicity is maintained by a durable, self-renewing memory T cell cluster within the CSF. This is especially important since experimental findings have been inconsistent, often providing limited support for a directly pathogenic role of these antibodies (2–7). Nevertheless, the authors showed the presence of numerous CSF BCRs with 25% mAbs being GAD65-reactive against nonhuman primate brain tissues as well as a small percentage being GAD65-reactive against Purkinje-cells.

By combining single-cell transcriptomics with paired immune-repertoire analysis in CSF and blood, Barman et al. argue for a compartmentalized, chronic CNS immune process centered on clonally expanded CD4⁺ stem cell-like memory T cells (TSCM), alongside somatically matured GAD65-reactive B cells in the CSF.

The work by Barman et al. implies sequential and coherent immunopathology: The association of expanded T cell clones with inflammatory, proliferative, and cytotoxic transcriptional programs further supports disease relevance rather than simple bystander activation. Of note, data were complemented by histological analyses from intraparenchymal tissue specimens, demonstrating presence of CD8⁺ TSCM in three of four anti-GAD AINS patients with rare meningeal and intraparenchymal CD4⁺ TSCM in one patient (Fig. 1).

This fits remarkably well with data about GAD65 neuropathology. Recent pathological work in GAD-associated temporal lobe epilepsy has shown an early inflammatory stage with cytotoxic T lymphocytes and plasma cells in afflicted tissue, together with intrathecal IgG synthesis, but without convincing evidence for classic complement-mediated antibody injury (8). Moreover, a portion of the cells identified had an antigen-specific resident memory T cell phenotype, fitting well with the data shown by Barman et al. Those data argued that neurons are more likely to be injured by cell-mediated immunity than by membrane attack complex deposition, potentially via CD8⁺ TSCM sitting in the brain parenchyma following putative activation by CD4⁺ TSCM in the CSF and meninges, as plausibly argued by Barman et al. The present work does not directly prove that the expanded CSF TSCM are GAD65-specific or neurotoxic, but it narrows the gap between pathology and liquid-biopsy neuroimmunology.

Indeed, the paper's stage-dependent pattern with peripheral plasmablasts early and TSCM enrichment later argues that anti-GAD65 autoimmunity may evolve from a more overt B cell phase into a more entrenched T cell-maintained phase. That model may have direct therapeutic implications, because it predicts that timing matters: B and plasma-cell directed strategies may be most useful early after disease onset, whereas chronic disease may require interventions capable of dismantling long-lived pathogenic T cell memory. Of note, immunotherapy in anti-GAD65 AINS immunotherapy is effective in a substantial proportion of patients, as shown in a cohort of 27 patients with 70% therapeutic response, yet without complete recovery in any of the patients (9).

In light of the findings by Barman et al. with crucial involvement of memory T cells, the recent clinical success in treating anti-GAD65 positive SPS-patients with anti-CD19 directed CAR T cells is only at first sight counterintuitive: a likely explanation for the apparent efficacy is that its benefit may extend beyond simple antibody reduction (10). By inducing deep depletion of B cells and plasmablasts, CAR T cells not only lower humoral immune activity but may also disrupt B cell-dependent antigen presentation, costimulation, and cytokine support that sustain autoreactive T cell responses. In this framework, clinical improvement does not necessarily imply that GAD65 antibodies are directly pathogenic (2); rather, it may reflect interruption of the broader B cell–T cell network that maintains chronic autoimmunity. This interpretation is consistent with the first published anti-CD19 CAR T cell case in severe treatment-refractory SPS, which reported marked clinical improve-

ment (11), a report showing benefit even in advanced refractory SPS and MG (12) and with the topline data from the phase 2 KYSA-8 trial of anti-CD19 CAR T cells in SPS, in which 26 treated patients showed a statistically significant improvement in the timed 25-foot walk at week 16 together with highly significant benefits across secondary disability and stiffness measures (13).

The current work by Barman et al. reframes the disease in a more coherent immunological picture: anti-GAD65 neurological syndromes appear to involve a compartmentalized, antigen-driven intrathecal response in which affinity-matured GAD65-reactive B cells coexist with clonally expanded, inflammatory T cell populations, particularly CD4⁺ TSCM in the CSF and meninges and CD8⁺ TSCM in the brain parenchyma. That framework can explain why antibodies are diagnostically indispensable yet therapeutically insufficient as a standalone explanation. It also helps to understand why some patients become chronically refractory: Long-lived stem-like T cell populations are exactly the kind of immune reservoir that conventional short-course immunotherapy or also B cell directed therapeutics such as anti-CD20 antibodies may fail to eradicate. As discussed by the authors approaches such as anti-CD3 directed therapeutic strategies, maybe supported with a “brain shuttle approach,” or tailored CAR T cell approaches might be a promising approach to target anti-GAD65 AINS in the future; potentially reducing risk of long-term neuronal damage and preventing disability accumulation.

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